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(54) **EDIBLE BEAN LINE <17589> (SHINE)**

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(57) **ABSTRACT**

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Related U.S. Application Data

(60) Provisional application No. 63/557,458, filed on Feb.
23, 2024.

An edible bean seed designated as <17589> or SHINE, a sample of the edible bean deposited under accession no. PI 705623 is disclosed. Methods of using the edible bean seed designated as <17589> or SHINE for breeding new varieties of bean seed are also disclosed, as well as seeds of the edible bean seed designated as <17589> or SHINE.

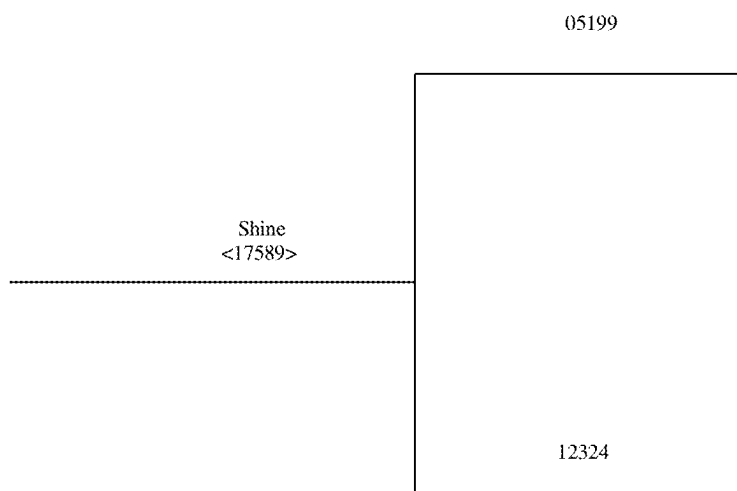


FIGURE 1

EDIBLE BEAN LINE <17589> (SHINE)**CROSS-REFERENCE TO RELATED APPLICATION**

[0001] This application claims priority to U.S. Provisional Patent Application No. 63/557,458, filed Feb. 23, 2024, the contents of the entirety of which is incorporated herein by this reference.

TECHNICAL FIELD

[0002] The present invention relates generally to the field of plant breeding, and more specifically to edible bean line <17589> (SHINE).

BACKGROUND OF THE INVENTION

[0003] In the United States, there is increasing concern over the consumption of high sugar and high fat foods by the population. An alarming increase in obesity among the citizens in the United States exists since many people in the United States are overweight due to a lack of exercise and poor eating habits.

[0004] Beans and peas are listed under both the Vegetables, and the Meats and Beans categories of the Dietary Guidelines Pyramid, thus, emphasizing the healthy nature of legumes in the diet. In addition to the emphasis on lower fat and lower sugar diets, beneficial effects of legumes are becoming apparent. Legumes are packed with fiber and protein, as well as being a good source of numerous vitamins and minerals. Legumes also have a low glycemic index and may help play a role in maintaining normal blood sugar levels; increasing digestive health; and even possibly improving heart health. Yet, legumes are an underutilized food in the United States diet.

[0005] With the increased awareness of the problems of obesity, poor eating habits, and the increased awareness of the health benefits of legumes, there exists a need for healthier food products and foodstuffs that provide the beneficial nutritional effects of legumes, as well as a consistent crop supply of such legumes.

SUMMARY OF THE INVENTION

[0006] In each of its various embodiments, the present invention helps fulfill these needs and discloses a pinto bean variety for use in breeding new bean varieties, as well as a foodstuff.

[0007] In one embodiment, an edible bean seed designated as SHINE, a sample of the edible bean seed deposited under accession no. PI 705623, on about Aug. 8, 2024, is disclosed.

[0008] In another embodiment, methods of using the edible bean seed SHINE for breeding new varieties of bean seed are disclosed. Seeds of the edible bean seed designated as SHINE are also disclosed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 shows the breeding history of edible bean line <17589> (SHINE).

DETAILED DESCRIPTION OF THE INVENTION

[0010] In one embodiment, an edible bean line designated as <17589> or (SHINE), is disclosed. As used herein, the term SHINE will be used to refer to edible bean variety

<17589>. Also disclosed are bean plants having the physiological and morphological characteristics of the edible bean line designated as SHINE. Parts of the bean plant including, without limitation, pollen, ovules, pods and cells, as well as uses of such parts in breeding are further disclosed.

[0011] In another embodiment, seed of the edible bean line designated as SHINE is disclosed. Such seed may be an essentially homogenous population of the edible bean line designated as SHINE, wherein such seed of the edible bean line designated as SHINE is essentially free of other seed. Accordingly, the seed of the present invention comprises at least 95% or more of the seed of the edible bean line designated as SHINE.

[0012] In yet a further embodiment, tissue culture of regenerable cells of the edible bean line designated as SHINE are disclosed. Such tissue culture may be capable of expressing all of the physiological and morphological characteristics of the edible bean line designated as SHINE and be capable of regenerating plants having the same genotype of plants of the edible bean line designated as SHINE. The tissue culture may be obtained from embryos, meristems, cotyledons, pollen, leaves, anthers, roots, root tips, pistil, flower, seed and/or stalks.

[0013] In an additional embodiment, breeding methods for producing edible bean seeds, plants and/or pods using the edible bean line designated as SHINE of the present invention as a parent are disclosed. Such breeding methods may be used to prepare hybrid edible bean seed or plants or inbred edible bean seed or plants, where seed ultimately produced using the edible bean line designated as SHINE as at least one of the parents are included within the scope of this invention.

[0014] In one embodiment, methods of crossing the edible bean line designated as SHINE with itself of a second plant and the seeds and plants produced by such methods are disclosed. Such methods may be used for propagating edible bean lines designated as SHINE, or can be used to produce hybrid dry bean seeds and the plants grown therefrom.

[0015] In a further embodiment, the present invention discloses a process for producing beans comprising obtaining a plant of the edible bean line designated as SHINE, cultivating the plant to maturity, and collecting and/or harvesting beans from the mature plant. Such beans may be used ultimately as a foodstuff, as seed or as parent in a breeding program.

[0016] The present invention further discloses methods and compositions relating to plants, seeds and derivatives of edible bean line designated as SHINE. SHINE has an erect growth habit, is indeterminate, has guides that are medium to long and has no ability to climb. SHINE also has scattered pods that are not concentrated high or low, is adapted to machine harvest, and a fair lodging resistance. SHINE has an average beak length of 13.2 mm. Observations and seed increase show that edible bean line designated as SHINE shows uniformity and stability, without variants.

Origin and Breeding History

[0017] The development of the edible bean line designated as SHINE can be summarized as follows. A cross was made between ProVita pinto variety '05199' as the female parent and ProVita slow darkening pinto breeding line '12324' as the male parent. The cross was made during the spring in Year 1 in the greenhouse. The edible bean line designated as SHINE was derived from this cross.

[0018] The breeding history of edible bean line designated as SHINE is shown in FIG. 1.

TABLE 1

The following Table indicates the selections used to produce the edible bean line designated as SHINE.				
Year	Location	Generation	Bulk	Single Plant Selections
Spring Year 1	Kuna, ID-greenhouse	F1	X	
Summer Year 1	Twin Falls, ID-field	F2	X	
Summer Year 2	Twin Falls, ID-field	F3		X
Spring Year 3	Kuna, ID-greenhouse	F4	X	
Summer Year 3	Twins Falls, ID-field	F5	X	
Summer Year 4	Twin Falls, ID-field	F6	X	
Summer Year 5	Twin Falls, ID-field	F7		X
Summer Year 6	Kuna, ID-greenhouse	F8	X	
Summer Year 7	Twin Falls, ID-field	F9	X	
Summer Year 8	Twin Falls, ID-field	F10		Commercial Seed Increase
Summer Year 9	Idaho fields	F11		Commercial Seed Increase

[0019] Selections were done in the F3 and F7 generations as follows. In the F3 generation, there was segregation for sd gene for slow darkening, architecture, maturity, seed gloss, seed color, yield and seed size. Single plant selections were made for sd gene, upright architecture, high pod set, yield, seed gloss, seed color, seed size, and acceptable maturity. In the F7, generation, single plant selections were made for upright architecture, high pod set, yield, seed color, seed size, and acceptable maturity. Such selections resulted in the new edible bean line designated as SHINE, which was observed to be uniform, stable, and substantially free of any variants within commercially acceptable limits since the F8 generation. It is known that a small percentage of off-types may occur for a characteristic of edible bean line designated as SHINE during multiplication.

Physiological and Morphological Characteristics of Edible Bean Line Designated as SHINE

[0020] Table 2 depicts the physiological and morphological characteristics of edible bean line designated as SHINE.

TABLE 2

Characteristic	Edible bean line designated as SHINE
Market Class	Pinto
Maturity	Medium (90-100 days)
Days from planting to harvest maturity	92
Days from planting to harvest maturity of check variety (UI-114)	84
Plant Habit	Erect growth habit-indeterminate, guides
Type	medium to long with no ability to climb

TABLE 2-continued

Characteristic	Edible bean line designated as SHINE
Average height of mature plant	Not measured
Pod position	Scattered (not concentrated high or low)
Adaptability to machine harvest	Adapted
Lodging resistance	Fair
Leaflet morphology	Smooth and semi-glossy
Leaflet Shape	Ovate
Apex of leaflet	Acuminate
Base of leaflet	Obtuse
Flower color and days to bloom	White
Color of standard	
Color of wings	White
Color of keel	White
Days to 50% bloom	49
Pod Morphology	Solid
Green color pattern	
Mature color pattern	Solid
Green primary color	Green
Mature primary color	Tan
Green color modifier	Light medium
Mature color modifier	Light medium
Green secondary color	Not determined
Mature secondary color	Not determined
Green cross section shape	Pear
Mature cross section shape	Round
Green pod curvature	Slightly curved
Mature pod curvature	Slightly curved
Green pod beak orientation	Curved downward
Mature pod beak orientation	Curved downward. 1.32 cm average beak length.
Green constrictions	Deep
Mature constrictions	Deep
Average number of seeds/pod	5.7
Seed Color	Semi-shiny and polychrome
Primary color	White
Secondary color	Brown
Color pattern	Mottled
Hilar ring	Present (faint)
Hilar ring color	Yellow
Seed Shape and Weight	Cuboid
Shape of seed taken from middle of pod	
Dry seed weight in g/100 g	40
Seeds (adjusted to 12% moisture)	
Anthocyanin Pigmentation*	Absent
Flowers	
Leaves	Absent
Stems	Absent
Petioles	Absent
Pods	Absent
Peduncles	Absent
Seeds	Present
Nodes	Absent

TABLE 2-continued

Characteristic	Edible bean line designated as SHINE
Known Disease Resistance	Intermediate resistance to bean rust (<i>Uromyces appendiculatus</i>) Resistant to Bean Common Mosaic Virus (BCMV), NL8

Deposit Information

[0021] A deposit of the edible bean line designated as SHINE of the present invention was made with the National Center for Genetic Resources Preservation, Fort Collins, Colorado. The date of deposit was about Aug. 8, 2024. Upon issuance of a patent, all restrictions upon the deposit will be removed, and the deposit is intended to meet all of the requirements of 37 C.F.R. § 1.801-1.809. The accession number for the deposited seeds of edible bean line designated as SHINE is PI 705623. The deposit will be maintained in the depository for a period of 30 years, or 5 years after the last request, or for the enforceable life of the patent, whichever is longer, and will be replaced if necessary, during such period.

[0022] The present invention has been described with reference to certain exemplary embodiments, legume products, compositions and uses thereof. However, it will be recognized by those of ordinary skill in the art that various substitutions, modifications, or combinations of any of the exemplary embodiments may be made without departing from the spirit and scope of the invention. Thus, the invention is not limited by the description of the exemplary embodiment, but rather by the appended claims as originally filed.

What is claimed is:

- 1. An edible bean seed designated as SHINE, a sample of the edible bean seed deposited under accession no. PI 705623.
- 2. A plant, or parts thereof, produced by growing the edible bean seed of claim 1.
- 3. Pollen of the plant of claim 2.

- 4. An edible bean plant, or parts thereof, having all of the physiological and morphological characteristics of the edible bean plant of claim 2.
- 5. A tissue culture of regenerable cells of the edible bean plant of claim 2.
- 6. The tissue culture of claim 5, wherein the cells are obtained from a plant part selected from the group consisting of leaf, pollen, embryo, meristematic cell, root, root tip, anther, stomatal cell, flower, seed, stem, pod and combinations of any thereof.
- 7. An edible bean plant regenerated from the tissue culture of claim 6, having all of the morphological and physiological characteristics of an edible bean plant grown from an edible bean seed designated as SHINE, a sample of the edible bean seed deposited under accession no. PI 705623.
- 8. A method for producing an edible bean seed comprising crossing two edible bean plants and harvesting the resulting edible bean seed, wherein at least one of the two edible bean plants is the edible bean plant of claim 2.
- 9. A method for producing a hybrid edible bean seed comprising crossing the edible bean plant of claim 2 with a second edible bean plant and harvesting the resulting hybrid edible bean seed.
- 10. A plurality of the edible bean seeds of claim 1.
- 11. A method of planting a field, comprising planting the plurality of the edible bean seeds of claim 10 in the field.
- 12. A process for producing edible beans for consumption, comprising processing the plurality of the edible bean seeds of claim 10 such that the plurality of the edible bean seeds are suitable for consumption.
- 13. A method for producing an edible bean seed designated as a SHINE-derived edible bean seed, comprising:
 - crossing a plant of edible bean designated as SHINE, a sample of the seed deposited under accession number PI 705623, with a second edible bean plant to yield progeny edible bean seed; and
 - growing the progeny edible bean seed to yield SHINE-derived edible bean plants and allowing seed of SHINE-derived edible beans plants to form.
- 14. The method of claim 13, wherein the second edible bean plant is transgenic.
- 15. An edible bean designated as SHINE-derived edible bean plant produced by the method of claim 13.

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